AMPHIREACH AR-Mini – HARDWARE CIRCUIT DIAGRAM

SENSORS

MLX90642-D45
thermal · forward 45°
3V3
GND
SDA0
SCL0

MLX90640-110°
thermal · cradle
3V3
GND
SDA1
SCL1

BNO055
9-DoF IMU
3V3
GND
SDA0
SCL0

LEAK PROBE
2 bare wires · hull floor
OUT
GND

VBAT SENSE
100k / 33k divider
ADC
GND

MAIN CONTROLLER

(XIAO ESP32S3 SENSE)

5V
GND
3V3
A0 VBAT
A1 LEAK/KILL
D2 SDA1
D3 SCL1
D4 SDA0
D5 SCL0

XIAO ESP32S3 SENSE
OV2640 · microSD · WiFi

D6 TX1
D7 RX1
USB-C
D8–D10 SPI→SD
GPIO21 SD CS

USB-C ↔ USB-A

FreeRTOS · core 0: perception (thermal blobs, pan loop) · core 1: control (state machine, motors, CSV log)

PWM DRIVER

(PCA9685 · I2C 0x40 · 16 ch)

3V3
GND
SDA
SCL
OE

PCA9685
16-ch 12-bit PWM

CH0
CH1
CH2
CH3
CH4
CH5
CH6
CH7–9

KILL → OE: всички PWM изходи off, хардуерно

THRUSTERS

THRUSTER L · CCW
THRUSTER R · CW
→ prop

TRACK DRIVERS

(BTS7960 ×2 · 43 A)

BTS7960 L
M+ M−

BTS7960 R
M+ M− → ROB0154 · 12 V · 160 rpm

INDICATORS & SAFETY

BUZZER
NPN BC337

LED ×3
state · arm · fault

KILL SW

5 V → BTS7960 VCC · R_EN · L_EN

SERVO BUS

(Waveshare Serial Bus Servo Driver · UART 1 Mbps)

BUS DRIVER
TTL↔servo bus

TX
RX
12V
GND

5264-3P daisy chain
30 cm + 90 cm cables

ST3215 ID1
cradle L

ST3215 ID2
cradle R

ST3215 ID3
cam pan

ST3215 ID4
dome v2

POWER SUPPLY

LiPo 3S 2200 mAh
XT60 · 40C

KILL SW 50 A

FUSE BLOCK
6-slot blade
20 A thrusters
15 A servo bus
10 A tracks
5 A logic
5 A Pi

12 V in

BUCK 12→5 V
3 A · logic rail

GND common

RASPBERRY PI 5 (v2: ROS 2 bridge · telemetry)

USB-C PWR
USB-A

RASPBERRY PI 5
8 GB · active cooler · 64 GB

BUCK 12→5 V 5 A
USB-C out · 25 W

ROB0154 chassis

GND

XIAO ↔ Pi over USB (serial) · ROS 2 bags · WiFi dashboard

LEGEND

+12 V
+5 V
3V3
GND
PWM / signal
I2C
UART / bus
Motor out

Силовите вериги (thruster-и, серво шина, вериги) идват директно от бушонния блок; през XIAO и PCA9685 минават само сигнали. Обща маса за всички модули. Kill: 50 A ключ на батерията (hardware) + OE на PCA9685 (логика) + ADC стълбица към A1.